

Interview Transcript with Dr. Felix Schneider (Cologne, Germany) Pulmonologist & Research Lead, University Hospital Cologne

Interviewer: Dr. Schneider, thank you for agreeing to this conversation. To start, could you tell us your background and how you got involved in the COVID response? Dr. Schneider: Thank you. I am Felix Schneider, a pulmonary physician and clinical researcher at the University Hospital in Cologne. Before the pandemic, much of my work focused on chronic lung disease COPD, fibrosis, asthma and clinical trials of novel therapies. When COVID-19 emerged, my group pivoted to study lung injury specific to SARS-CoV-2, and I also took on a leadership role in our hospital's COVID respiratory care teams.

We were fortunate to have experience in ARDS and ventilator research, which gave us a baseline. But COVID challenged nearly every assumption we had about respiratory failure. I found myself splitting time between the lab and the wards.

Interviewer: How did your hospital's pulmonary division adapt when COVID surged? Dr. Schneider: Cologne's hospital network expanded ICU capacity rapidly; our pulmonary department repurposed wards as intermediate respiratory care units (step-downs). We deployed high-flow nasal oxygen and noninvasive ventilation earlier, trying to prevent intubation where possible, while monitoring closely for signs of deterioration.

We also reorganized our research labs to support COVID: tissue banking, measurement of biomarkers, and imaging studies. Patients admitted with COVID often had serial CT scans to study disease progression. That research-fed-back-into-clinical practice was extremely stressful yet deeply motivating.

Interviewer: What unique pulmonary challenges did COVID pose in your experience? Dr. Schneider: COVID lung injury has peculiar features. Many patients had compliance preservation lungs that could receive volume but had severe shunting and hypoxia. That required very nuanced ventilation strategies. We also saw rapid progression in some cases, with microvascular involvement capillary thrombosis, endothelial damage which traditional ARDS models did not fully explain.

Another challenge was post-COVID fibrosis. Some patients, months after recovery, showed radiological features of fibrotic change. We're still researching which patients develop permanent scarring and how to intervene early.

Interviewer: Did your research group identify any biomarkers or predictors of outcome? Dr. Schneider: Yes, we tracked markers like surfactant protein D (SP-D), KL-6, and soluble RAGE, correlating them with CT progression and outcomes. One finding was that patients whose SP-D failed to decrease in the first week had higher odds of progression to mechanical ventilation.

We also analyzed cytokine profiles and found that a subset of patients had hyperinflammatory signatures resembling macrophage activation syndrome. That led us to trial more aggressive

immunomodulation in select cases.

Interviewer: How did you balance research and clinical duties under pressure? Dr. Schneider: It was brutal. Some days I would round in the ICU from 7 a.m. to 4 p.m., then head to the lab to process samples until midnight. Sleep was intermittent. But the urgency made everything feel necessary. When you see a patient deteriorate while you think about a cytokine curve, you want your lab work to inform the next patient's care.

Over time, I had to accept that not all research would be published immediately; some was strictly to guide bedside decisions in real time. That tension between ideal science and urgent care was one of the hardest parts.

Interviewer: Germany's approach emphasized evidence-based adaptation. How did that influence pulmonary care here? Dr. Schneider: We continuously revised our protocols. Early on, some advocated high PEEP strategies; later, we refined to best PEEP for COVID lungs rather than blanket high pressures. Prone positioning was adopted more liberally, even earlier in moderate ARDS. Noninvasive therapies were used more aggressively, balanced by close surveillance.

Importantly, we didn't adopt every new therapy—we waited for data. For example, ivermectin and unproven antivirals were rejected in our protocols until sufficient evidence emerged. That cautious, data-first approach prevented wasted resources and harm.

Interviewer: How did staff burnout and mental health factor into your leadership? Dr.

Schneider: It was significant. Many pulmonary staff lived in hotel rooms to protect families, and every shift felt existential. As leaders, we instituted rotating rest weeks, mental health check-ins, and peer support. I also tried to lead by example—I would go into the wards, help move lines, talk with staff. That visibility matters. Burnout is reduced when people feel seen.

We also took time to mark small victories—discharges, extubations, recoveries—with team acknowledgments. Emotional sustenance came from reminding ourselves why we do this.

Interviewer: What role did vaccination and variant tracking play in your pulmonary practice?

Dr. Schneider: Vaccination dramatically reduced severe respiratory cases over time. As vaccine uptake increased, the average severity of lung disease among hospitalized patients declined, and fewer required extended ventilation. That shift allowed us to repurpose beds for non-COVID pulmonary emergencies.

Variant tracking was important for anticipating therapy responsiveness. For example, some variants seemed associated with greater inflammatory lung injury, which adjusted our immunomodulation thresholds. We collaborated with virology labs in Cologne to sequence viral isolates from our cohort and correlated that with disease phenotype.

Interviewer: Could you share a case or moment that profoundly affected you? Dr. Schneider: One patient, a 55-year-old with no prior lung disease, developed progressive respiratory failure. Over five days, despite maximal support, his CT scans showed bilateral consolidation. We

escalated care to ECMO. He eventually recovered but had months of persistent dyspnea and required rehabilitation. He later returned to thank the staff – tears were shed on all sides.

That case reminded me that lung disease is not just about survival – it's about quality of life, reconditioning, and hope in the long term.

Interviewer: How do you view Germany's pulmonary research integration now, post-peak waves? Dr.

Schneider: The integration has strengthened. Pulmonary divisions, critical care, virology, and imaging now have ongoing collaborative pipelines. We're designing long-term cohorts of recovered patients to study lung repair. In terms of infrastructure, funding for translational lung research has increased post-COVID – not as a temporary response, but as a commitment.

We also now have protocols ready to activate for future respiratory pandemics – a sleeper research-clinical infrastructure that can scale rapidly.

Interviewer: What were Finland's or Scandinavian hospital lessons you observed (given proximity)? Dr. Schneider: Some Scandinavian centers emphasized home-care oxygen and telemonitoring early on – sending moderate patients home with pulse oximeters and monitoring rather than admitting them prematurely. That reduced hospital overload. We adopted similar practices in Cologne's peripheral clinics.

They also integrated primary care with pulmonology more seamlessly, enabling early referrals and outpatient follow-ups, which reduced chronic damage. That holistic approach – not just hospital-centric – taught us to think broader than ICU.

Interviewer: From your vantage, what are the most important lessons for pulmonary medicine going forward? Dr. Schneider: A few: – Always anticipate vascular and endothelial components, not just alveolar injury. – Biomarker-guided care has real value – use it to stratify patients rather than treat all equally. – Post-COVID lung care must become standard pulmonary fibrosis, rehabilitation, and long-term follow-up can't be afterthoughts. – Invest in clinician-researchers – bridging lab and bedside accelerates insight.

Also, respiratory pandemics will recur – we must treat readiness not as a project, but as a duty.

Interviewer: What has this experience taught you personally about medicine? Dr. Schneider: It deepened my humility. No matter how experienced, a novel pathogen is a humbling teacher. I realized that beyond protocols, patients remember your voice, presence, and empathy.

It also showed me that research and care are two sides of the same coin. You heal partly by understanding, and understand partly by treating. That synergy is not optional – it's essential.

Interviewer: Thank you, Dr. Schneider, for your honesty, insight, and work. Dr. Schneider:

Thank you. May these stories guide us not only in crisis, but in strengthening everyday medicine.

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